CC Assignment

Satya Prakash
2329391601 7

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1) Architecture of a Helm chart

⇒ A Helm chart is a package used to define install & manage kubernetes application. its architecture consists of following main components

- Chart.Yaml – Contains metadata such as the chart name version & description
- Values.Yaml – Stores default configuration values used in templates.
- Templates/ – Contains kubernetes resource definitions written as templates.
- Charts/ – Includes dependent charts (Subcharts)
- Readme.md – Provides documentation about chart.

These components together make depolyment resusable & easy to manage.

2) Advantages & limitations of Serverless (Faas)

⇒ Advantages	Limitations
• No need to manage servers	• Cold start latency issues
• Automatic scaling based on demand	• Limited execution time
• Pay-as-you use cost model	• Vendor lock in risk
• Faster development & deployment	• Difficult debugging & monitoring.

3) Google Big Query handles large-scale analytics efficiently using:

→ Distributed architecture : Process data across multiple nodes.

→ Columnar Storage : Reads only required colums improving performance

→ Parallel processing : Executes queries simultaneously for faster results

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→ Serverless model - No need to manage infrastructure

→ Dremel execution engine - Enables high speed query processing.

(4) Comparison :-

Feature	Prebuilt ML APIs	AutoML	Custom ML Models
Expertise Req	Low	Medium	High
Flexibility	Low	Medium	High
Development time	Fast	Moderate	Slow
Control	Limited	Partial	Full

* Prebuilt ML APIs : Ready to use solutions for tasks like vision or speech
* AutoML : Allows model creation with minimal coding effort
* Custom ML Models : Fully customizable but requires deep expertise.

(5) The lifecycles of serverless function includes

(i) Trigger - An event occurs (HTTP req, file upload)

(ii) Execution - The function runs in response to the trigger

(iii) Scaling - Automatically scales up or down based on demand.

(iv) Termination - Stops execution after completing the task.

This ensures efficient resource utilization.

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6) Role of Helm in DevOps & CI/CD pipelines
→ Simplifies deployment of Kubernetes applications
→ Enables version control of application configurations
→ Supports automation in CI/CD pipelines
→ Allows easy rollback to previous versions
→ Ensures consistency across different environments
• Helms improves efficiency & reliablity in DevOps workflows.

7) Working of edge inference: it refers to running machine models on edge devices instead of cloud.
Steps :-
* Data is collected on the edge device
* Data is preprocessed locally
* A trained model is deployed on the device
* Real time inference is performed
* Immediate output is generated.
This reduces latency & enhances data privacy.